Supplementary Material to: "A Multidimensional Agent-Based Modeling and Simulation Framework for Urban Water Demand Forecast under Socioeconomic Inequalit"

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Abstract

Traditional aggregate models fail to capture complex interactions between spatial heterogeneity and human behavior in urban water demand forecasting. We present a spatially explicit Agent-Based Model for Water Prediction (ABM-WP) framework to simulate neighborhood-scale water consumption dynamics on the GAMA platform with data from Salvador, Brazil. A key contribution is the systematic analysis of multiple scenarios with 18 configurations integrating the Environmentalist, Moderate, and Wasteful behavioral profiles with socioeconomic income growth and population expansion. ABM-WP was compared to statistical Seasonal ARIMA and machine learning models (LSTM, MLP, SVM). ANOVA and Tukey tests revealed significant tipping points where socioeconomic catalysts cause demand to deviate from historical trends over 220,000 m^3 annually. ABM-WP identifies critical peaks that data-based methods fail to predict. Spatial consumption patterns analysis is driven by connection density and socioeconomic clustering. ABM-WP is a robust decision-support tool for governance, providing insights for urban sustainability targeting conservation and infrastructure planning.

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[1] Research conducted while a visiting professor at the Federal University of Bahia (UFBA).

1. **PEAS (Performance Measures, Environment, Actuators, and Sensors) to describe the task environment of the AA, PA, and CA agents.**

Table 1: PEAS description for the AA, responsible for data collection, preprocessing, and analysis. Source: Elaborated by the author.

PEAS	Description
Performance	Quantity and quality of data processed per consumer unit
Environment	Virtual with georeferenced addresses. Classified as Stochastic (unpredictable events), Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Analyze family-grouped data Compare and process consumption Send processed data to predictor agent
Sensors	Residence geolocation Residence types and profile Population growth

Table 2: PEAS description for the PA, responsible for making scenario simulations. Source: Elaborated by the author.

PEAS	Description
Performance	Maintain prediction quality with minimal Mean Absolute Percentage Error (MAPE)
Environment	Virtual with georeferenced addresses. Classified as Stochastic (unpredictable events), Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Select optimal prediction model Generate family consumption forecasts Send forecasts to AC
Sensors	Data provided by AA

Table 3: PEAS description for the CA, responsible for consolidating the generated forecasts. Source: Elaborated by the author.

PEAS	Description
Performance	Number of reports with consumption forecasts issued
Environment	Virtual with georeferenced addresses. Classified as Stochastic (unpredictable events), Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Send water demand reports Distribute to PA water demand reports
Sensors	Data generated by the PA

2. Modules

The ABM-WP architecture applies reactive agent types (percept/act) with a vertically layered structure. The architecture is structured into four-layer modules: Interface, Data Normailzation, Agents, and Reports.

- Interface - focuses on the role of data input providers, actors responsible for manual data collection, and the web interface that receives and stores multiple data inputs.
- Data Normalization - currently responsible for processing location and housing type data, but scalable as needed by the project.
- Agent - handles communication between the involved agents and the flow of exchanges based on their responsibilities.
- Reporting - responsible for generating reports based on the analyses performed by the system using the provided data.

The inference mechanism of the ABMS-WP framework is executed through a hybrid approach. An inductive component is employed during the initial calibration phase, in which reference consumption patterns are extracted from the last 12 months of historical records. Subsequently, a deductive approach is applied through conditional rules that guide the behavior of agent across different socioeconomic stressors.

The transition of water demand over time is modeled as a recursive function where the state of an agent i at time $t+1$ depends on its previous state adjusted by demographic and socio-economic drivers. The general inference rule is defined by Eq. 1

$$C_{i,t+1} = C_{i,t} \times F_{pop}(S_{pop}, B_i) \times F_{income}(S_{inc}, I_{i,t}) \tag{1}$$

where:

- $C_{i,t+1}$ is the predicted consumption;
- $C_{i,t}$ is the consumption from the previous period;
- F_{pop} is the population growth function determined by the scenario S_{pop} (uniform or random) and the agent neighborhood context B_i;
- F_{income} is the income adjustment function based on the economic scenario S_{inc}; and
- current income level of the agent $I_{i,t}$.

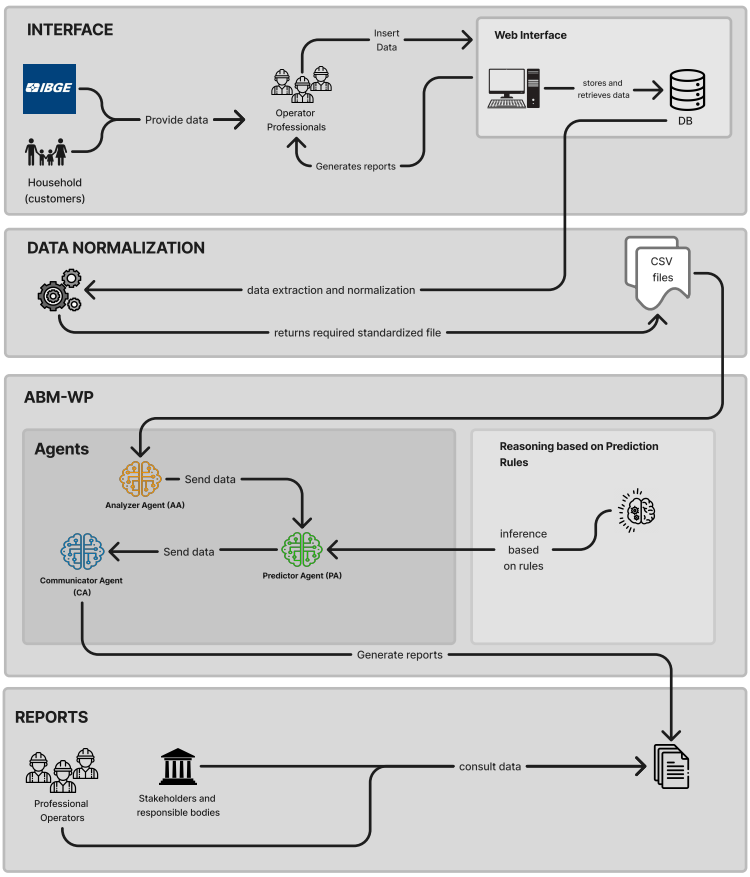

Figure 1: Architecture of the ABM-WP framework, encompassing the interaction between actors, tasks and agents.

3. Arquitetural Desgin

The architectural model fo ABM-WP framework, was structured under the Tropos methodology, details the interaction between human actors and the agents responsible for water demand forecast. The Operator acts as the central point of progression, managing data dependencies with the Family (data provider) and the Meter (responsible for recording monthly and historical consumption). Operational intelligence is distributed across three specialized agents: the Analyzer Agent, focused on analyzing, preprocessing, and consolidating cus-

tomer data; the Predictor Agent, responsible for generating individualized consumption forecasts per family and per scenario; and the Communicator Agent, which prepares and consolidates scenario predictions. The workflow is driven by dependencies and objectives.

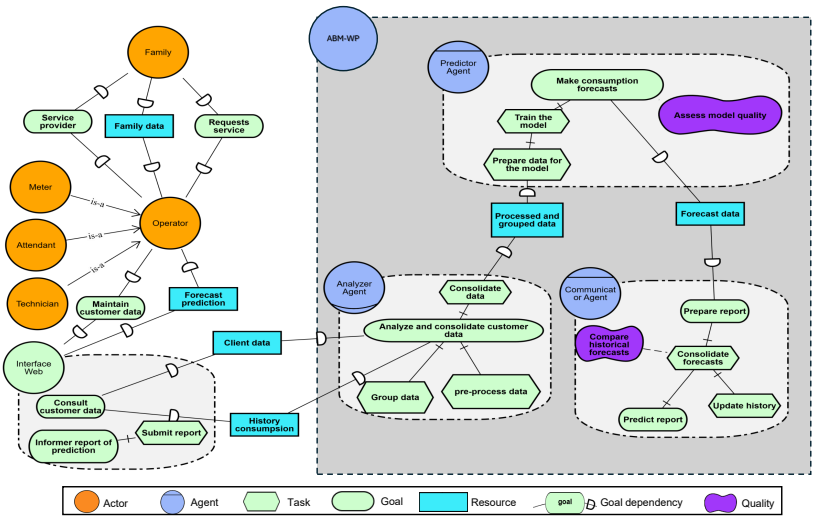

Figure 2: Architecture of the ABM-WP framework, encompassing the interaction between actors, tasks and agents.

4. List of consumption profiles by sector

Ref.: January/2026

CD_SETOR	total_consumers	environmentalist_count	moderate_count	wasteful_count	avg_daily_consumption	pct_environmentalist	pct_moderate	pct_wasteful
S027	658	314	76	268	145.80	47.72	11.55	40.73
S029	582	319	49	214	154.32	54.81	8.42	36.77
S023	554	316	38	200	131.63	57.04	6.86	36.10
S048	489	296	54	139	104.16	60.53	11.04	28.43
S003	453	292	33	128	104.80	64.46	7.28	28.26
S024	445	248	30	167	135.99	55.73	6.74	37.53
S020	413	206	30	177	141.14	49.88	7.26	42.86
S078	403	220	43	140	125.34	54.59	10.67	34.74
S079	394	223	35	136	107.66	56.60	8.88	34.52
S032	354	220	40	94	104.59	62.15	11.30	26.55
S099	351	152	37	162	147.69	43.30	10.54	46.15
S080	348	206	49	93	91.65	59.20	14.08	26.72
S067	343	211	28	104	111.11	61.52	8.16	30.32
S021	334	224	34	76	85.51	67.07	10.18	22.75
S000	323	232	25	66	77.44	71.83	7.74	20.43
S008	302	72	23	207	188.08	23.84	7.62	68.54
S026	300	195	25	80	96.62	65.00	8.33	26.67
S086	295	158	37	100	130.44	53.56	12.54	33.90
S063	274	153	22	99	117.77	55.84	8.03	36.13
S033	273	151	26	96	113.87	55.31	9.52	35.16
S007	272	78	18	176	160.73	28.68	6.62	64.71
S034	268	125	27	116	168.65	46.64	10.07	43.28
S005	267	195	22	50	85.48	73.03	8.24	18.73
S025	254	152	21	81	119.07	59.84	8.27	31.89
S073	238	154	21	63	87.61	64.71	8.82	26.47
S081	234	143	17	74	104.89	61.11	7.26	31.62
S093	228	178	9	41	61.78	78.07	3.95	17.98
S074	222	152	19	51	97.31	68.47	8.56	22.97
S077	215	97	22	96	223.85	45.12	10.23	44.65
S092	210	125	7	78	121.78	59.52	3.33	37.14
S028	209	120	19	70	244.10	57.42	9.09	33.49
S018	190	140	7	43	66.58	73.68	3.68	22.63
S072	188	65	17	106	315.92	34.57	9.04	56.38
S084	186	108	13	65	115.18	58.06	6.99	34.95

Ref.: January/2026

CD_SETOR	total_consumers	environmentalist_count	moderate_count	wasteful_count	avg_daily_consumption	pct_environmentalist	pct_moderate	pct_wasteful
S022	183	80	17	86	150.74	43.72	9.29	46.99
S060	173	64	20	89	230.03	36.99	11.56	51.45
S066	173	108	15	50	104.44	62.43	8.67	28.90
S082	173	125	14	34	76.27	72.25	8.09	19.65
S019	172	73	19	80	175.34	42.44	11.05	46.51
S083	163	116	13	34	83.41	71.17	7.98	20.86
S004	162	104	12	46	93.19	64.20	7.41	28.40
S036	160	76	16	68	182.12	47.50	10.00	42.50
S047	158	104	15	39	87.81	65.82	9.49	24.68
S089	157	112	12	33	87.37	71.34	7.64	21.02
S015	153	99	15	39	93.24	64.71	9.80	25.49
S017	151	111	10	30	65.02	73.51	6.62	19.87
S001	146	67	21	58	124.41	45.89	14.38	39.73
S044	138	43	14	81	353.31	31.16	10.14	58.70
S006	138	92	16	30	85.21	66.67	11.59	21.74
S091	136	59	15	62	171.26	43.38	11.03	45.59
S031	127	43	8	76	389.42	33.86	6.30	59.84
S075	118	46	14	58	155.11	38.98	11.86	49.15
S035	105	59	8	38	132.70	56.19	7.62	36.19
S030	103	27	6	70	278.62	26.21	5.83	67.96
S045	97	21	8	68	347.00	21.65	8.25	70.10
S068	91	21	2	68	445.09	23.08	2.20	74.73
S095	87	60	13	14	77.27	68.97	14.94	16.09
S039	83	32	10	41	144.35	38.55	12.05	49.40
S088	83	39	8	36	157.17	46.99	9.64	43.37
S046	79	35	6	38	210.04	44.30	7.59	48.10
S094	77	51	6	20	89.96	66.23	7.79	25.97
S040	66	24	5	37	210.93	36.36	7.58	56.06
S013	23	11	2	10	134.02	47.83	8.70	43.48
S002	1	0	0	1	71778.69	0.00	0.00	100.00